

AI in Genomics 2026

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GOHREP it! Interactive discussions

From a big biological question to a clear research plan

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GOHREP it!

From a big biological question to a clear
research plan



Sophien Kamoun ✓

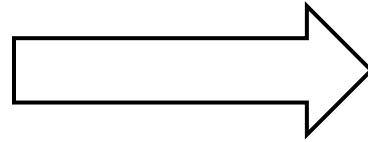
@KamounLab

You cannot do research without a BBQ—Big Biological Question

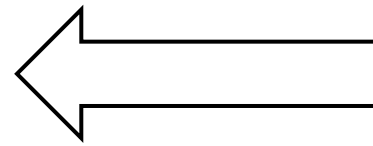
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***You cannot do research
without a BBQ!***



Lots of tools, lots of data



See the big picture

Why it's critical

- **It's the filtering mechanism** – Most ideas don't deserve pursuit.
- **It's the communication currency** – Audiences don't engage with a list of methods.
- **It sustains across a multi-year project.** Why does it matter will always be relevant.
- **It gives hypotheses meaning** – The BBQ is what the hypothesis is in service of.
- **It forces resource prioritisation.**
“Is this worth the resource? Is it in service of my BBQ?”
- **It's the direct antidote to tool-first thinking.**
“Does this tool serve your BBQ, or are you just running it?”

Your BBQs – what works

How do plant-associated microorganisms influence plant responses to environmental stress and contribute to plant resilience?

- Amira Yagoubi (INSAT) - Amazing example!

- It's ambitious and exciting
- Sufficiently specific – captures research niche
- Not too specific it can be answered with a single experiment
- Not too specific that a general audience cannot understand it
- Tool agnostic
- It's enduring



Your BBQs – to avoid

Addressing too many questions at once

Tool-centric (not future proofed)

Too vague or too specific

Mistakenly addressing an intermediary question rather than a BBQ (journey vs destination)



Address your BBQ GOHREP it!

Goal

Hypothesis

Rationale

Experimental Plan

Disposable, revised, and often discarded, not fixed at the outset.

GOHREP

1. **Goal:** This is the main objective of the research project—what you ultimately aim to achieve. It should be specific enough to provide clear direction but not overly detailed.
2. **Hypothesis:** An educated guess or statement of fact that the research intends to test. A well-defined hypothesis is central to scientific inquiry and should be specific and testable.
3. **Rationale:** The logical reasoning behind the hypothesis. This section describes the observations or preliminary data that led to the formulation of the hypothesis.
4. **Experimental Plan:** The detailed approach and specific experiments designed to test the hypothesis. This plan outlines the methods and procedures that will be used to challenge the hypothesis with the goal of achieving the research objective.

Build a GoHREP live

GOHREP: can AlphaFold tell you two proteins interact?

GOal

Decide which of three candidate pathogen proteins interacts with Pikp-1, using AlphaFold 3 confidence metrics alone.

Hypothesis

Only one of the three jobs will show confidence in how the two chains are placed relative to each other. The other two return complete models that score low.

Rationale

- AF3 folds every job it is given. A returned complex is not evidence of a complex.
- pLDDT and pTM mostly describe the chains. ipTM is the score that describes the interface.

Experimental plan

1. Submit three two chain jobs, the same receptor each time.
2. Record pTM and ipTM. Check whether pTM separates them at all.
3. Read the off-diagonal block of each PAE plot.
4. Rank by ipTM, then look at where the chains touch in the top model.

Discuss your BBQ

What is my BBQ?

Discuss the BBQ with colleagues.

Broadly sketch your GoHREP?

Is it structurally sound?

Is in service of my BBQ?

Goal

Hypothesis

GoHREP canvas

Rationale

Experimental Plan

“park-it” mechanism

What is park-it?

Connect the technical detail with your research.

Why park-it?

The cognitive-load problem.

Learning how a tool works and simultaneously mapping it onto your own system are two demanding tasks, competing for the same working memory.

How to park-it?

Offload the thought instead of chasing it.

Create a GoHREP canvas and keep it open. When a "this connects to my project" thought lands, drop a one-line note in the relevant box and return to the session

Don't fall into the trap

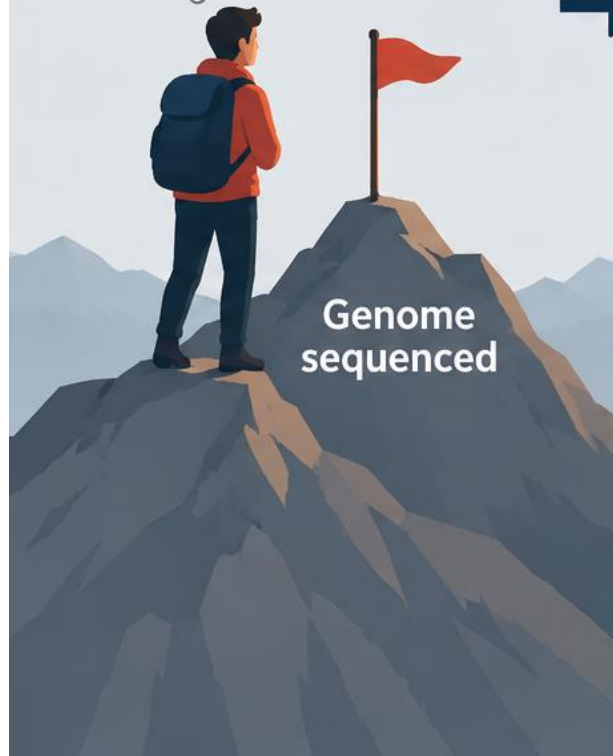
Like any data, a prediction is evidence, not a conclusion

AlphaFold doesn't answer a biological question

A summit isn't the end – it's the beginning

Some see the summit as the goal.

We got the genome.
We're done!



In reality, the summit gives you a wider view
– new peaks and new questions.

Understand
disease
mechanisms



Discover
new drugs
and therapies



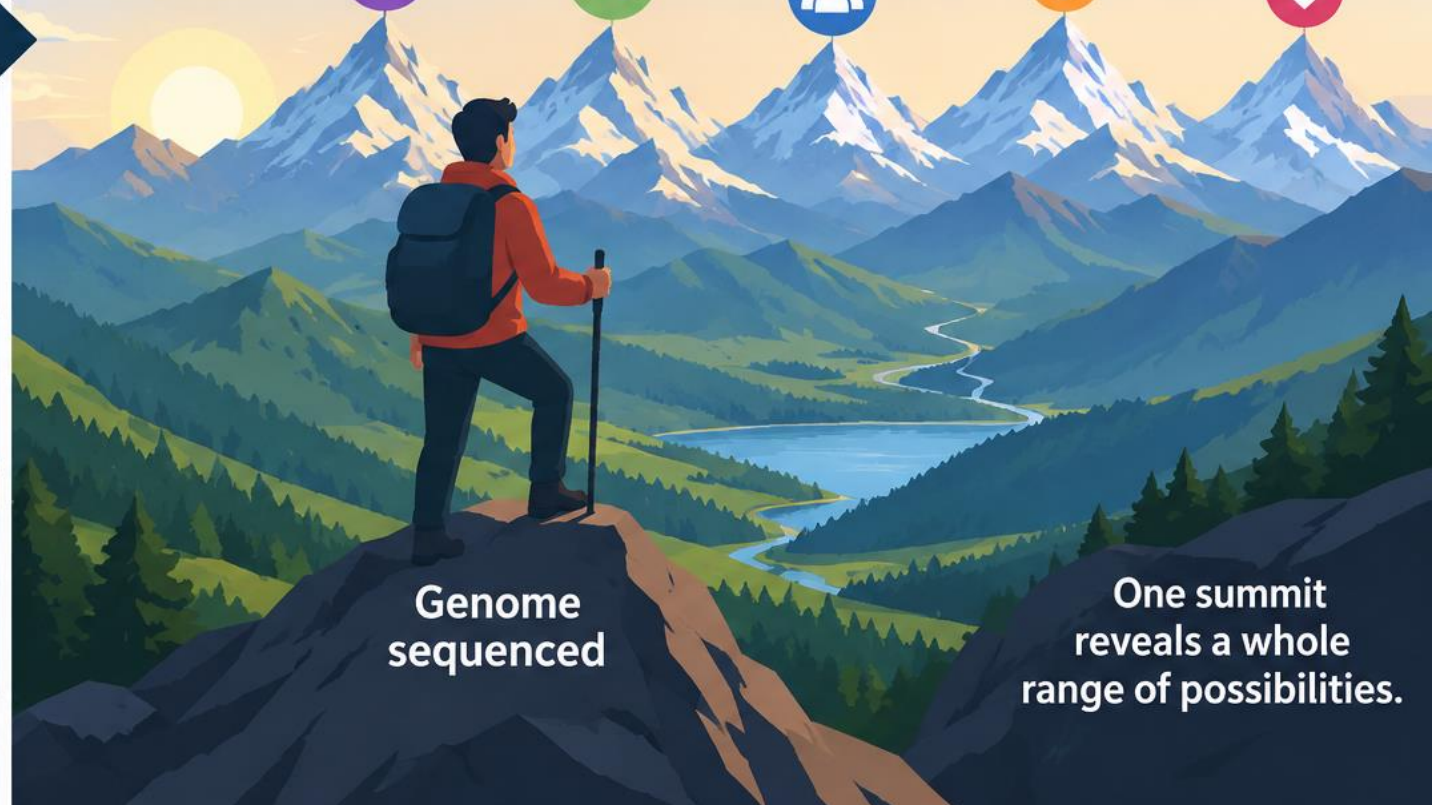
Predict
individual
risk



Explore
evolution
and diversity



Enable
healthier
futures



Genome
sequenced

One summit
reveals a whole
range of possibilities.

Programme

The same two-day programme runs at both venues (Espace Zmorda, then ISBS).

Day 1 Predict & Interpret

08:30–09:30	Registration
09:30–09:45	Welcome & Introductions
09:45–10:15	Introduction to AI for Genomics
10:15–11:00	Theoretical Introduction to AlphaFold
11:00–11:20	Break
11:20–12:30	 Practical 1: Predict
12:30–13:30	Lunch
13:30–15:00	<i>GOHREP it!</i> Interactive discussions
15:00–15:15	Break
15:15–17:00	 Practical 2: Interpret
17:00–17:15	Q&A

Day 2 Compare & Design

09:00–09:30	Day 2 Welcome & Registration
09:30–11:15	 Practical 3: Compare
11:15–11:30	Break
11:30–12:30	Model your own proteins
12:30–13:30	Lunch
13:30–14:45	 Practical 4: Design
14:45–15:05	Break
15:05–16:15	Genomics in an AI world
16:15–16:30	Summary
16:30–17:00	Q&A
17:00	Meeting close